NYCU Introduction to Machine Learning, Homework 1

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Part. 1, Coding (60%):

(10%) Linear Regression Model - Closed-form Solution

1. (10%) Show the weights and intercepts of your linear model.

```
2025-10-07 19:10:57.296 | INFO     | __main__:main:99 - LR_CF.weights=array([ 7.3935217 , 17.59620753,  0.80104512,  0.56381241])
, LR_CF.intercept=55.2859
```

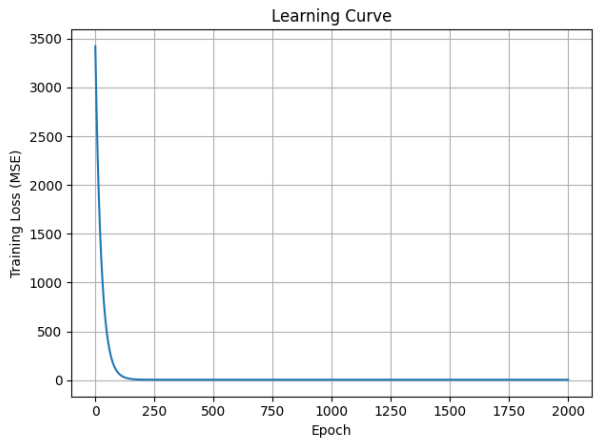
(40%) Linear Regression Model - Gradient Descent Solution

2. (10%)
   - Show the hyperparameters of your setting (e.g., learning rate, number of epochs, batch size, etc.).
   - Show the weights and intercepts of your linear model.

```
class LinearRegressionGradientdescent(LinearRegressionBase):
    def fit(self, X, y, learning_rate=1e-2, epochs=2000, decay=1e-4):
```

```
2025-10-07 21:54:09.170 | INFO     | __main__:main:111 - LR_GD.weights=array([ 7.3935217 , 17.59620753,  0.80104512,  0.56381241])
, LR_GD.intercept=55.2859
```

3. (10%) Plot the learning curve. (x-axis=epoch, y-axis=training loss)



4. (20%) Show your MSE.cf, MSE.gd, and error rate between your closed-form solution and the gradient descent solution.

```
2025-10-07 21:59:40.691 | INFO     | __main__:main:127 - Prediction difference: 0.0000
2025-10-07 21:59:40.691 | INFO     | __main__:main:132 - mse_cf=4.2903, mse_gd=4.2903. Difference: 0.000%
```

(10%) Code Check and Verification

5. (10%) Lint the code and show the PyTest results.

```
(.venv) PS C:\Users\matth\OneDrive\桌面\機器學習\hw1> flake8 main.py
(.venv) PS C:\Users\matth\OneDrive\桌面\機器學習\hw1>
```

```
(.venv) PS C:\Users\matth\OneDrive\桌面\機器學習\hw1> pytest -v -s test_main.py
>>
========================================================= test session starts =========================================================
platform win32 -- Python 3.8.5, pytest-8.3.5, pluggy-1.5.0 -- c:\users\matth\onedrive\桌面\機器學習\.venv\scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\matth\OneDrive\桌面\機器學習\hw1
collected 2 items

test_main.py::test_regression_cf (100, 1)
2025-10-07 22:55:16.175 | INFO     | test_main:test_regression_cf:30 - model.weights=array([[3.]]), model.intercept=array([4.])
PASSED
test_main.py::test_regression_gd (100, 1)
2025-10-07 22:55:16.183 | INFO     | main:fit:66 - EPOCH 0, loss=30622.0606, lr=0.000100
2025-10-07 22:55:16.579 | INFO     | main:fit:66 - EPOCH 10000, loss=0.9997, lr=0.000050
2025-10-07 22:55:16.936 | INFO     | main:fit:66 - EPOCH 20000, loss=0.1975, lr=0.000033
2025-10-07 22:55:17.285 | INFO     | main:fit:66 - EPOCH 30000, loss=0.0625, lr=0.000025
2025-10-07 22:55:17.569 | INFO     | main:fit:66 - EPOCH 40000, loss=0.0256, lr=0.000020
2025-10-07 22:55:17.815 | INFO     | main:fit:66 - EPOCH 50000, loss=0.0123, lr=0.000017
2025-10-07 22:55:18.070 | INFO     | main:fit:66 - EPOCH 60000, loss=0.0067, lr=0.000014
2025-10-07 22:55:18.335 | INFO     | test_main:test_regression_gd:44 - model.weights=array([3.]), model.intercept=3.93751640563475
3
PASSED
```

Part. 2, Questions (40%):

1. (10%) Linear models $y = w^\top x + b$ have limited fitting power.
   a. In one sentence, explain why a single linear model is limited.
   b. Give one concrete task that a single linear model cannot solve, and state why no single hyperplane/affine function solves it.

   a. The model $y = w^\top x + b$ can only produce a single hyperplane as its decision boundary, so it cannot separate data with nonlinear or multi-region patterns.
   b. Task: Classify points inside a circle as class 1, and points outside as class 0.
   Reason: A single linear model cannot classify circular data (inside vs. outside a circle) because the decision boundary is curved, not a flat hyperplane.

2. (15%) Why do we add a regularization term in linear regression? What are the differences between L2 regularization (Ridge) and L1 regularization (Lasso)? <u>Please explain in detail</u>.

   Regularization penalizes large weights so that the model does not overfit by excessively adjusting parameters to fit every training sample, leading to better stability and generalization.
   L2 Regularization (Ridge): λ∑w^2，It shrinks all coefficients toward zero but does not make them exactly zero. Thus, Ridge keeps all features in the model but reduces their influence.
   L1 Regularization (Lasso): λ∑ | w |，It can force some coefficients to be exactly zero, performing feature selection. This makes the model sparse (uses only a subset of features).

3. (15%)
   - What is overfitting? Under what conditions can a model overfit? (List two) How can overfitting be alleviated? (List two)

1.

Overfitting happens when a model learns not only the underlying pattern in the training data but also the noise or random fluctuations, resulting in poor performance on test data or unseen data.

2.

a. The model is too complex: too many parameters or high-degree polynomial.

b. Insufficient or noisy data: too few samples, or the data contains random errors.

3.

a. Add regularization: L1 or L2, as mentioned above.

b. Use more data or apply data augmentation to better represent the true distribution.